

## Abhishek Halder

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| CONTACT INFORMATION       | Howe 2341<br>Department of Aerospace Engineering<br>Iowa State University, Ames, IA 50011 USA                                                                                                                                                                                                                                                                                           | +1-979-583-6070<br>ahalder@iastate.edu<br>abhishekhalder.org |
| PROFESSIONAL APPOINTMENTS | <b>Associate Professor</b><br><br>Department of Aerospace Engineering<br>Member: Translational AI Center<br>Iowa State University                                                                                                                                                                                                                                                       | July 2023 – Current                                          |
|                           | <b>Associate Adjunct Professor</b><br><br>Department of Applied Mathematics<br>University of California, Santa Cruz                                                                                                                                                                                                                                                                     | July 2023 – Current                                          |
|                           | <b>Faculty Scholar</b><br><br>Physics Division<br>Lawrence Livermore National Lab                                                                                                                                                                                                                                                                                                       | January 2025 – April 2025                                    |
|                           | <b>Assistant Professor</b><br><br>Department of Applied Mathematics<br>Department of Electrical and Computer Engineering (affiliated faculty)<br>University of California, Santa Cruz                                                                                                                                                                                                   | October 2017 – June 2023                                     |
|                           | <b>Postdoctoral Scholar</b><br><br>Department of Mechanical and Aerospace Engineering<br>University of California, Irvine<br>– Topic: <i>Stochastic control, filtering and optimal transport</i><br>– Host: Tryphon T. Georgiou                                                                                                                                                         | February 2017 – September 2017                               |
|                           | <b>Postdoctoral Research Associate</b><br><br>Department of Electrical and Computer Engineering<br>Texas A&M University<br>– Topic: <i>Demand response in smart grid, unmanned aerial systems traffic management</i><br>– Host: P.R. Kumar                                                                                                                                              | June 2014 – January 2017                                     |
|                           | <b>Global Research Innovation and Technology Intern</b><br><br>Systems Modeling and Control<br>Eaton Corporation, Eden Prairie, Minnesota<br>– Topic: <i>Voltage Control in UPS: Modeling, Controller Synthesis, and Hardware-in-loop Simulation</i><br>– Host: Yigang Wang<br>– Projects: (1) Model Based Design in Power Electronics<br>(2) Co-operative Control of Generator and UPS | Summer 2013                                                  |

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|                       | <b>Visiting Researcher</b>                                                                                                                                                                                                                                                                                                                                                                                        | Summer 2012 |
|                       | Dynamics and Control, Coordinated Science Laboratory<br>University of Illinois at Urbana-Champaign <ul style="list-style-type: none"> <li>– Topic: <i>Nonlinear Estimation as Gradient Flow</i></li> <li>– Host: Prashant Mehta</li> </ul>                                                                                                                                                                        |             |
|                       | <b>Research Intern</b>                                                                                                                                                                                                                                                                                                                                                                                            | Summer 2006 |
|                       | Advanced Systems Development Section, Control Systems Group<br>Indian Space Research Organization Satellite Center (ISAC), Bangalore INDIA <ul style="list-style-type: none"> <li>– Topic: <i>A Study of Petri Nets: Modeling, Analysis and Simulation</i></li> <li>– Host: A. Venkateswarlu, Deputy Director, CSG, ISAC</li> </ul>                                                                               |             |
| EDUCATION             | <b>Ph.D. in Aerospace Engineering</b>                                                                                                                                                                                                                                                                                                                                                                             | May 2014    |
|                       | <b>Texas A&amp;M University</b> , College Station, Texas USA <ul style="list-style-type: none"> <li>• Dissertation: <i>Probabilistic Methods for Model Validation</i><br/>(<b>Outstanding Doctoral Student Award</b>)</li> <li>• Advisor: Raktim Bhattacharya</li> </ul>                                                                                                                                          |             |
|                       | <b>Bachelors and Masters in Aerospace Engineering</b>                                                                                                                                                                                                                                                                                                                                                             | July 2008   |
|                       | <b>Indian Institute of Technology Kharagpur</b> , West Bengal INDIA <ul style="list-style-type: none"> <li>• Thesis: <i>Development of An Autonomous Reconfigurable UAV</i><br/>(<b>Best Dual Degree Thesis Award</b>)</li> <li>• Advisor: Manoranjan Sinha</li> </ul>                                                                                                                                            |             |
| RESEARCH<br>INTERESTS | <b>Broad area</b><br>Systems, control, optimization, ML/AI                                                                                                                                                                                                                                                                                                                                                        |             |
|                       | <b>Focus areas</b> <ul style="list-style-type: none"> <li>– Mathematical foundation of generative AI</li> <li>– Stochastic control</li> <li>– Optimal transport</li> <li>– Schrödinger bridge</li> <li>– Diffusion models</li> </ul>                                                                                                                                                                              |             |
| GRANTS                | <b>G9. <i>Advancing the Theory and Algorithms for Distribution Control: Nonlinearity, Interaction, Robustness.</i></b><br>PI: Abhishek Halder [Iowa State University], Co-PI: Yongxin Chen [Georgia Tech].<br>Duration: 08/01/2025–07/31/2028.<br><b>National Science Foundation.</b><br>Total award amount: \$611,720.<br>Abhishek’s award amount: \$313,839.<br>Indirect costs for Abhishek’s budget: \$87,214. |             |
|                       | <b>G8. <i>REU Supplement: Computation-Aware Algorithmic Design for Cyber-Physical Systems.</i></b><br>PI: Abhishek Halder [Iowa State University].<br>Duration: 09/01/2024–12/31/2024.<br><b>National Science Foundation.</b>                                                                                                                                                                                     |             |

Abhishek's award amount: \$10,000.  
Indirect costs for Abhishek's budget: \$0.

**G7.** *Computation-Aware Algorithmic Design for Cyber-Physical Systems.*

PI: Ricardo Sanfelice [UC Santa Cruz], Co-PIs: Abhishek Halder [UC Santa Cruz], Heiner Litz [UC Santa Cruz], Murat Arcak [UC Berkeley], Linh Pham [UPenn], Jonathan Sprinkle [Vanderbilt University], Majid Zamani [University of Colorado at Boulder].

Duration: 07/01/2022–07/30/2027.

**National Science Foundation.**

Total award amount: \$5.78M.

Abhishek's award amount: \$685,735.

Indirect costs for Abhishek's budget: \$208,699.

**G6.** *Learning and Distributional Feedback Control for Fabrication of Advanced Materials.*

PI: Abhishek Halder [UC Santa Cruz], Co-PI: Ali Mesbah [UC Berkeley].

Duration: 08/01/2021–07/31/2025.

**National Science Foundation.**

Total award amount: \$650,461.

Abhishek's award amount: \$296,069.

Indirect costs for Abhishek's budget: \$80,549.

**G5.** *Optimal Transport Algorithms for Stochastic Uncertainty Management in Modern Power Systems.*

PI: Abhishek Halder [UC Santa Cruz].

Duration: 09/01/2019–08/31/2022.

**National Science Foundation.**

Total/Abhishek's award amount: \$279,830.

Indirect costs for Abhishek's budget: \$52,813.

**G4.** *Computation-Aware Algorithmic Design for Cyber-Physical Systems for Intelligent Transportation.*

PI: Ricardo Sanfelice [UC Santa Cruz], co-PIs: Heiner Litz [UC Santa Cruz], Abhishek Halder [UC Santa Cruz].

Duration: 09/01/2020–09/30/2021.

**UCSC Office of Research Seed Grant.**

Total award amount: \$75,000.

Abhishek's award amount: \$0.

Indirect costs for Abhishek's budget: \$22,057.

**G3.** *Fast Computation of Stochastic Reachability for Provably Safe Planning and Control in Automated Driving.*

Lead PI: Abhishek Halder [UC Santa Cruz], Ford PI: Baljeet Singh [Ford].

Duration: 01/01/2018–12/31/2019.

**University Research Project, Ford Motor Company.**

Total/Abhishek's award amount: \$50,000.

Indirect costs for Abhishek's budget: \$0.

**G2.** *Cloud-based Anytime Computation of Reachable Tubes for Provably Safe Unmanned Aerial Systems Traffic Management.*

Lead PI: Abhishek Halder [UC Santa Cruz], Co-I: Mark Mueller [UC Berkeley], Ricardo Sanfelice [UC Santa Cruz], Claire Tomlin [UC Berkeley].

Duration: 04/01/2018–06/30/2019.

**Center for Information Technology Research in the Interest of Society**

**(CITRIS) Seed Fund Award.**

Total award amount: \$60,000.

Abhishek's award amount: \$30,000.

Indirect costs for Abhishek's budget: \$0.

**G1. Real-time Computation of Reachable Tubes for Provably Safe Unmanned Aerial Systems Traffic Management.**

PI: Abhishek Halder [UC Santa Cruz].

Duration: 02/01/2018–08/31/2018.

**UCSC Committee on Research New Faculty Research Grant.**

Total/Abhishek's award amount: \$4,000.

Indirect costs for Abhishek's budget: \$0.

JOURNAL  
PUBLICATIONS

**J40.** S. Shivakumar, G.A. Bondar, G. Khan, and **A. Halder**. Sum-of-Squares Programming for Ma-Trudinger-Wang Regularity of Optimal Transport Maps. *Journal of Optimization Theory and Applications*, 2026.

doi:

**J39.** I. Nodot, D. Gligorijevic, and **A. Halder**. Learning Concave Bid Shading Strategies in Online Auctions via Measure-valued Proximal Optimization. *IEEE Control Systems Letters*, 2025.

doi:10.1109/LCSYS.2025.3640064

**J38.** G.A. Bondar, and **A. Halder**. Optimal Multimarginal Schrödinger Bridge: Minimum Spanning Tree over Measure-valued Vertices. *IEEE Control Systems Letters*, 2025.

doi:10.1109/LCSYS.2025.3634944

**J37.** A.M.H. Teter, W. Wang, and **A. Halder**. Schrödinger Bridge with Quadratic State Cost is Exactly Solvable. *IEEE Transactions on Automatic Control*, 2025.

doi:10.1109/TAC.2025.3631521

**J36.** A.M.H. Teter, **A. Halder**, M.D. Schneider, A.S. Perloff, J. Pratt, C.M. Artman, and M. Demireva. Control-affine Schrödinger Bridge and Generalized Bohm Potential. *IEEE Control Systems Letters*, 2025.

doi:10.1109/LCSYS.2025.3623945

**J35.** A. Eisenklam, R. Gifford, G.A. Bondar, Y. Cai, T. Sial, L.T.X. Phan, and **A. Halder**. Rasco: Resource Allocation and Scheduling Co-design for DAG Applications on Multicore. *ACM Transactions on Embedded Computing Systems*, 2025.

doi:10.1145/3761814

**J34.** **A. Halder**. [Book Review] Optimal Transport: A Comprehensive Introduction to Modeling, Analysis, Simulation, Applications. *IEEE Control Systems Magazine*, 2025.

doi:10.1109/MCS.2025.3573816

**J33.** G.A. Bondar, R. Gifford, L.T.X. Phan, and **A. Halder**. Stochastic Learning of Computational Resource Usage as Graph Structured Multimarginal Schrödinger Bridge. *IEEE Transactions on Control Systems Technology*, 2025.

doi:10.1109/TCST.2025.3560511

**J32.** A.M.H. Teter, I. Nodot, and **A. Halder**. Probabilistic Lambert's Problem: Connections with Optimal Mass Transport, Schrödinger Bridge and Reaction-Diffusion PDEs. *SIAM Journal on Applied Dynamical Systems*, 2024.

doi:10.1137/24M1646145

- J31.** S. Haddad, P. Khodary, and **A. Halder**. Exact Computation of LTI Reach Set from Integrator Reach Set with Bounded Input. *IEEE Control Systems Letters*, 2023. doi:10.1109/LCSYS.2023.3347173
- J30.** A.M. Teter, I. Nodozi, and **A. Halder**. Proximal Mean Field Learning in Shallow Neural Networks. *Transactions on Machine Learning Research*, 2023. <https://openreview.net/forum?id=vyRBsqj5iG>
- J29.** I. Nodozi, C. Yan, M. Khare, **A. Halder**, and A. Mesbah. Neural Schrödinger Bridge with Sinkhorn Losses: Application to Data-driven Minimum Effort Control of Colloidal Self-assembly. *IEEE Transactions on Control Systems Technology*, 2023. doi:10.1109/TCST.2023.3337588
- J28.** A.M. Teter, Y. Chen, and **A. Halder**. On the Contraction Coefficient of the Schrödinger Bridge for Stochastic Linear Systems. *IEEE Control Systems Letters*, 2023. doi:10.1109/LCSYS.2023.3326836
- J27.** S. Haddad, and **A. Halder**. A Note on the Hausdorff Distance between Norm Balls and their Linear Maps. *Set-Valued and Variational Analysis*, 2023. doi:10.1007/s11228-023-00692-1
- J26.** I. Nodozi, **A. Halder**, and I. Matei. A Controlled Mean Field Model for Chiplet Population Dynamics. *IEEE Control Systems Letters*, 2023. doi:10.1109/LCSYS.2023.3282174
- J25.** S. Haddad, and **A. Halder**. The Curious Case of Integrator Reach Sets, Part I: Basic Theory. *IEEE Transactions on Automatic Control*, 2023. doi:10.1109/TAC.2023.3244694
- J24.** **A. Halder**, K.F. Caluya, P. Ojaghi, and X. Geng. Stochastic Uncertainty Propagation in Power System Dynamics using Measure-valued Proximal Recursions. *IEEE Transactions on Power Systems*, 2022. doi:10.1109/TPWRS.2022.3217267
- J23.** S. Haddad, and **A. Halder**. Certifying the Intersection of Reach Sets of Integrator Agents with Set-valued Input Uncertainties. *IEEE Control Systems Letters*, 2022. doi:10.1109/LCSYS.2022.3179666
- J22.** S. Haddad, **A. Halder**, and B. Singh. Density-based Stochastic Reachability Computation for Occupancy Prediction in Automated Driving. *IEEE Transactions on Control Systems Technology*, 2022. doi:10.1109/TCST.2022.3145976
- J21.** K.F. Caluya, and **A. Halder**. Wasserstein Proximal Algorithms for the Schrödinger Bridge Problem: Density Control with Nonlinear drift. *IEEE Transactions on Automatic Control*, 2021. doi:10.1109/TAC.2021.3060704
- J20.** S. Haddad, K.F. Caluya, **A. Halder**, and B. Singh. Prediction and Optimal Feedback Steering of Probability Density Functions for Safe Automated Driving. *IEEE Control Systems Letters*, 5(6):2168–2173, 2021. doi:10.1109/LCSYS.2020.3045105
- J19.** **A. Halder**. Smallest Ellipsoid Containing  $p$ -Sum of Ellipsoids with Application to Reachability Analysis. *IEEE Transactions on Automatic Control*, 66(6):2512–2525, 2021. doi:10.1109/TAC.2020.3009036

- J18. A. Halder**, K.F. Caluya, B. Travacca, and S.J. Moura. Hopfield Neural Network Flow: A Geometric Viewpoint. *IEEE Transactions on Neural Networks and Learning Systems*, 31(11): 4869–4880, 2020.  
doi:10.1109/TNNLS.2019.2958556
- J17.** K.F. Caluya, and **A. Halder**. Gradient Flow Algorithms for Density Propagation in Stochastic Systems. *IEEE Transactions on Automatic Control*, 65(10):3991–4004, 2020.  
doi:10.1109/TAC.2019.2951348
- J16.** F.A.C.C. Fontes, **A. Halder**, J. Becerril, and P.R. Kumar, Optimal Control of Thermostatic Loads for Planning Aggregate Consumption: Characterization of Solution and Explicit Strategies. *IEEE Control Systems Letters*, 3(4):877–882, 2019.  
doi:10.1109/LCSYS.2019.2918978
- J15.** Z. Askarzadeh, R. Fu, **A. Halder**, Y. Chen, and T.T. Georgiou. Stability Theory for Stochastic Models in Opinion Dynamics. *IEEE Transactions on Automatic Control*, 65(2):522–533, 2020.  
doi:10.1109/TAC.2019.2912490
- J14. A. Halder**. DeGroot-Friedkin Map in Opinion Dynamics is Mirror Descent. *IEEE Control Systems Letters*. 3(2):463–468, 2019.  
doi:10.1109/LCSYS.2019.2900452
- J13. A. Halder**, and R.G. Sanfelice. CPAR Control Theory and Automation Symposium. *IEEE Control Systems Magazine*. February, 2019.  
doi:10.1109/MCS.2018.2876965
- J12. A. Halder**, X. Geng, F.A.C.C. Fontes, P.R. Kumar, and L. Xie. Optimal Power Consumption for Demand Response of Thermostatically Controlled Loads. *Optimal Control Applications and Methods*. 40(1):68–84, 2019.  
doi:10.1002/oca.2467
- J11. A. Halder**, X. Geng, P.R. Kumar, and L. Xie. Architecture and Algorithms for Privacy Preserving Thermal Inertial Load Management by A Load Serving Entity. *IEEE Transactions on Power Systems*. 32(4):3275–3286, 2017. [Paper selected by the IEEE Power & Energy Society (PES) Technical Committee for presentation in 2017 PES General Meeting.]  
doi:10.1109/TPWRS.2016.2628055
- J10. A. Halder**, K. Lee, and R. Bhattacharya. A Probabilistic Method for Nonlinear Robustness Analysis of F-16 Controllers. *Journal of Guidance, Control, and Dynamics*. 38(10):1935–1946, 2015.  
doi:10.2514/1.G000386
- J9.** P. Dutta, **A. Halder**, and R. Bhattacharya. Nonlinear Estimation with Perron-Frobenius Operator and Karhunen-Loève Expansion. *IEEE Transactions on Aerospace and Electronic Systems*. 51(4):3210–3225, 2015.  
doi:10.1109/TAES.2015.140591
- J8.** K. Lee, **A. Halder**, and R. Bhattacharya. Performance and Robustness Analysis of Stochastic Jump Linear Systems using Wasserstein Metric. *Automatica*. 51:341–347, 2015.  
doi:10.1016/j.automatica.2014.10.080
- J7. A. Halder**, and R. Bhattacharya. Probabilistic Model Validation for Uncertain Nonlinear Systems. *Automatica*. 50(8):2038–2050, 2014.  
doi:10.1016/j.automatica.2014.05.026

- J6.** T. Kalmár-Nagy, P. Wahi, and **A. Halder**. Dynamics of a Hysteretic Relay Oscillator with Periodic Forcing. *SIAM Journal on Applied Dynamical Systems*, 10(2):403–422, 2011.  
doi:10.1137/100784606
- J5.** **A. Halder**, and R. Bhattacharya. Dispersion Analysis in Hypersonic Flight During Planetary Entry Using Stochastic Liouville Equation. *Journal of Guidance, Control and Dynamics*, 34(2):459–474, 2011.  
doi:10.2514/1.51196
- J4.** S. Ghosh, **A. Halder**, and M. Sinha. Micro Air Vehicle Path Planning in Fuzzy Quadtree Framework. *Applied Soft Computing*, 11(8):4859–4865, 2011.  
doi:10.1016/j.asoc.2011.06.014
- J3.** S. Zhao, **A. Halder**, and T. Kalmár-Nagy. Nonlinear Dynamics of Unicycles in Leader-Follower Formation. *Communications in Nonlinear Science and Numerical Simulations*, 14(12):4204–4219, 2009.  
doi:10.1016/j.cnsns.2009.02.028
- J2.** S. Chauhan, C. Patil, M. Sinha, and **A. Halder**. Fuzzy State Noise Driven Kalman Filter for Sensor Fusion. *Journal of Aerospace Engineering, Proceedings of the Institution of Mechanical Engineers, Part G*, 223(8):1091–1097, 2009.  
doi:10.1243/09544100JAERO536
- J1.** **A. Halder**, R. Garhwal, V. Agarwal, and M. Sinha. Determination of Inertial Characteristics of A High Wing Unmanned Air Vehicle. *Journal of Institute of Engineers (India)*, 89:3–8, 2008.
- C45.** G.A. Bondar, A. Eisenklam, Y. Cai, R. Gifford, T. Sial, L.T.X. Phan, and **A. Halder**. Generative Profiling for Soft Real-Time Systems and its Applications to Resource Allocation. *32<sup>nd</sup> IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS 2026)*, Saint Malo, France, 2026. [Last five year average acceptance rate: 25.83%, source: <https://openaccept.org/c/sys/rtas/>]  
doi:
- C44.** K. Elamvazhuthi, and **A. Halder**. The Ground Cost for Optimal Transport of Angular Velocity. **Invited Paper**, *64<sup>th</sup> IEEE Conference on Decision and Control*, Rio de Janiero, Brazil, 2025.  
doi:10.1109/CDC57313.2025.11312415
- C43.** A.M.H. Teter, W. Wang, and **A. Halder**. Weyl Calculus and Exactly Solvable Schrödinger Bridges with Quadratic State Cost. **Invited Paper**, *60th Annual Allerton Conference*, Urbana, IL, 2024.  
doi:10.1109/Allerton63246.2024.10735300
- C42.** A.M.H. Teter, I. Nodozi, and **A. Halder**. Solution of the Probabilistic Lambert’s Problem: Optimal Transport Approach. **Invited Paper**, *26th International Symposium on Mathematical Theory of Networks and Systems (MTNS 2024)*, Cambridge, UK, 2024.  
doi:10.48550/arXiv.2402.01209
- C41.** G.A. Bondar, R. Gifford, L.T.X. Phan, and **A. Halder**. Path Structured Multimarginal Schrödinger Bridge for Probabilistic Learning of Hardware Resource Usage by Control Software. *American Control Conference*, Toronto, Canada, 2024.  
doi:10.23919/ACC60939.2024.10644864



- C40.** C. Yan, I. Nodozi, and **A. Halder**. Optimal Mass Transport over the Euler Equation. **Invited Paper**, *62<sup>nd</sup> IEEE Conference on Decision and Control*, Singapore, 2023.  
doi:10.1109/CDC49753.2023.10383425
- C39.** S. Haddad, and **A. Halder**. Convex and Nonconvex Sublinear Regression with Application to Data-driven Learning of Reach Sets. *American Control Conference*, San Diego, 2023.  
doi:10.23919/ACC55779.2023.10156204
- C38.** I. Nodozi, J. O’Leary, A. Mesbah, and **A. Halder**. A Physics-informed Deep Learning Approach for Minimum Effort Stochastic Control of Colloidal Self-Assembly. *American Control Conference*, San Diego, 2023.  
doi:10.23919/ACC55779.2023.10156176
- C37.** I. Nodozi, and **A. Halder**. Schrödinger Meets Kuramoto via Feynman-Kac: Minimum Effort Distribution Steering for Noisy Nonuniform Kuramoto Oscillators. *61<sup>st</sup> IEEE Conference on Decision and Control*, Cancún, Mexico, 2022.  
doi:10.1109/CDC51059.2022.9993420
- C36.** I. Nodozi, and **A. Halder**. A Distributed Algorithm for Measure-valued Optimization with Additive Objective. **Invited Paper**, *25th International Symposium on Mathematical Theory of Networks and Systems (MTNS 2022)*, Beyreuth, Germany, 2022.  
doi:10.48550/arXiv.2202.08930
- C35.** S. Haddad, and **A. Halder**. Boundary and Taxonomy of Integrator Reach Sets. *American Control Conference*, Atlanta, 2022.  
doi:10.23919/ACC53348.2022.9867312
- C34.** I.M. Balci, **A. Halder**, and E. Bakolas. On the Convexity of Discrete Time Covariance Steering in Stochastic Linear Systems with Wasserstein Terminal Cost. *60<sup>th</sup> IEEE Conference on Decision and Control*, Austin, 2021.  
doi:10.1109/CDC45484.2021.9683514
- C33.** S. Haddad, and **A. Halder**. Anytime Ellipsoidal Over-approximation of Forward Reach Sets of Uncertain Linear Systems. *Workshop on Computation-Aware Algorithmic Design of Cyber-Physical Systems*, CPS-IoT Week, 2021.  
doi:10.1145/3457335.3461711
- C32.** K.F. Caluya, and **A. Halder**. Reflected Schrödinger Bridge: Density Control with Path Constraints. *American Control Conference*, New Orleans, 2021.  
doi:10.23919/ACC50511.2021.9482813
- C31.** S. Haddad, and **A. Halder**. The Convex Geometry of Integrator Reach Sets. *American Control Conference*, Philadelphia, 2020.  
doi:10.23919/ACC45564.2020.9147611
- C30.** K.F. Caluya, and **A. Halder**. Finite Horizon Density Steering for Multi-input State Feedback Linearizable Systems. *American Control Conference*, Philadelphia, 2020.  
doi:10.23919/ACC45564.2020.9147847
- C29.** **A. Halder**, and T.T. Georgiou, Proximal Recursion for the Wonham Filter. **Invited Paper**, *58<sup>th</sup> IEEE Conference on Decision and Control*, Nice, France, 2019.  
doi:10.1109/CDC40024.2019.9030018
- C28.** K.F. Caluya, and **A. Halder**. Proximal Recursion for Solving the Fokker-Planck Equation. *American Control Conference*, Philadelphia, 2019.  
doi:10.23919/ACC.2019.8814363



- C27.** Z. Askarzadeh, R. Fu, **A. Halder**, Y. Chen, and T.T. Georgiou. Opinion Dynamics over Influence Networks. *American Control Conference*, Philadelphia, 2019.  
doi:10.23919/ACC.2019.8815341
- C26.** **A. Halder**. On the Parameterized Computation of Minimum Volume Outer Ellipsoid of Minkowski Sum of Ellipsoids. *57<sup>th</sup> IEEE Conference on Decision and Control*, Miami, 2018.  
doi:10.1109/CDC.2018.8619508
- C25.** **A. Halder**, and T.T. Georgiou. Gradient Flows in Filtering and Fisher-Rao Geometry. **Invited Paper**, *American Control Conference*, Milwaukee, 2018.  
doi:10.23919/ACC.2018.8431003
- C24.** **A. Halder**, and T.T. Georgiou. Gradient Flows in Uncertainty Propagation and Filtering of Linear Gaussian Systems. *56<sup>th</sup> IEEE Conference on Decision and Control*, Melbourne, 2017.  
doi:10.1109/CDC.2017.8264109
- C23.** **A. Halder**, and E.D.B. Wendel. Finite Horizon Linear Quadratic Gaussian Density Regulator with Wasserstein Terminal Cost. *American Control Conference*, Boston, 2016.  
doi:10.1109/ACC.2016.7526817
- C22.** **A. Halder**, X. Geng, G. Sharma, L. Xie, and P.R. Kumar. A Control System Framework for Privacy Preserving Demand Response of Thermal Inertial Loads. *IEEE International Conference on Smart Grid Communications (SmartGridComm 2015)*, Miami, 2015, pp. 181–186.  
doi:10.1109/SmartGridComm.2015.7436297
- C21.** **A. Halder**, K. Lee, and R. Bhattacharya. A Dynamical System Pair with Identical First Two Moments But Different Probability Densities. **Invited Paper**, *53<sup>rd</sup> IEEE Conference on Decision and Control*, Los Angeles, 2014.  
doi:10.1109/CDC.2014.7040335
- C20.** **A. Halder**, and R. Bhattacharya. Geodesic Density Tracking with Applications to Data Driven Modeling. **Invited Paper**, *American Control Conference*, Portland, 2014.  
doi:10.1109/ACC.2014.6859361
- C19.** K. Lee, **A. Halder**, and R. Bhattacharya. Probabilistic Robustness Analysis of Stochastic Jump Linear Systems. *American Control Conference*, Portland, 2014.  
doi:10.1109/ACC.2014.6859432
- C18.** **A. Halder**, and R. Bhattacharya. Frequency Domain Model Validation in Wasserstein Metric. *American Control Conference*, Washington DC, 2013.  
doi:10.1109/ACC.2013.6580754
- C17.** **A. Halder**, K. Lee, and R. Bhattacharya. Probabilistic Robustness Analysis of F-16 Controller Performance: An Optimal Transport Approach. *American Control Conference*, Washington DC, 2013.  
doi:10.1109/ACC.2013.6580708
- C16.** P. Dutta, **A. Halder**, and R. Bhattacharya. Nonlinear Filtering with Transfer Operator. *American Control Conference*, Washington DC, 2013.  
doi:10.1109/ACC.2013.6580302

- C15.** A. Halder, and R. Bhattacharya. Further Results on Probabilistic Model Validation in Wasserstein Metric. *51<sup>st</sup> IEEE Conference on Decision and Control (CDC)*, Maui, Dec. 2012.  
doi:10.1109/CDC.2012.6425987
- C14.** P. Dutta, A. Halder, and R. Bhattacharya. Uncertainty Quantification for Stochastic Nonlinear Systems with Perron-Frobenius Operator and Karhunen-Loève Expansion. *IEEE Multi-Conference on Systems and Control*, Dubrovnik, Croatia, Oct. 2012.  
doi:10.1109/CCA.2012.6402455
- C13.** A. Halder, and R. Bhattacharya. Model Validation: A Probabilistic Formulation. *50<sup>th</sup> IEEE Conference on Decision and Control (CDC) and European Control Conference (ECC)*, Orlando, Dec. 2011.  
doi:10.1109/CDC.2011.6161465
- C12.** A. Halder, and R. Bhattacharya. Beyond Monte Carlo: A Computational Framework for Uncertainty Propagation in Planetary Entry, Descent and Landing. *AIAA Guidance, Navigation and Control Conference*, Toronto, Aug. 2010.  
doi:10.2514/6.2010-8029
- C11.** S. Zhao, A. Halder, and T. Kalmár-Nagy. Leader-Follower Dynamics for Unicycles. *American Control Conference*, St. Louis, June 2009.  
doi:10.1109/ACC.2009.5160706
- C10.** S. Zhao, A. Halder, and T. Kalmár-Nagy. Nonlinear Dynamics of Unicycles in Leader-Follower Formation. *8th MSU-UAB Conference on Differential Equations and Computational Simulations*, Mississippi State University, May 2009.
- C9.** T. Kalmár-Nagy, A. Halder, and S. Zhao. Delay Tuned Phase Locking in A Pair of Coupled Limit Cycle Oscillators. *6<sup>th</sup> International Conference on Mathematical Modeling*, Vienna, Feb. 2009.
- C8.** S. Chauhan, C. Patil, A. Halder, and M. Sinha. FLIER: A Novel Sensor Fusion Algorithm. *3<sup>rd</sup> IEEE International Conference on Industrial and Information Systems*, IIT Kharagpur, Dec. 2008.  
doi:10.1109/ICIINFS.2008.4798459
- C7.** M. Sinha, A. Halder, R. Garhwal, N. S. Gopinath, and N. K. Malik. Lunar Satellite Observation Vector Construction using Non-rotating Origin and IAU2000A Precession-Nutation Model. *Conference on Advances in Space Science and Technology*, IIT Kharagpur, Jan. 2008.
- C6.** M. Sinha, A. Halder, R. Garhwal, A. K. Ghosh, N. S. Gopinath, and N. K. Malik. Lunar Gravity Field Modeling: A Critical Survey. *Conference on Advances in Space Science and Technology*, IIT Kharagpur, Jan. 2008.
- C5.** V. Agarwal, A. Halder, R. Garhwal, A. Gupta, S. Ghosh, S. Saxena, and M. Sinha. Inertial Characterization of Unmanned Aerial Vehicle AX-1. *4<sup>th</sup> International Conference on Theoretical, Applied, Computational and Experimental Mechanics*, IIT Kharagpur, Dec. 2007.
- C4.** A. Halder, S. Ghosh, and M. Sinha. Fuzzy Quadtree based Path Planner and Trajectory Smoother for A Low Cost Unmanned Aerial Vehicle. *3<sup>rd</sup> Indian International Conference on Artificial Intelligence*, Pune, Dec. 2007.
- C3.** R. Garhwal, A. Halder, and M. Sinha. Sensitivity Analysis using Neural Network for Estimating Aircraft Stability and Control Derivatives. *IEEE International Conference*

on *Intelligent and Advanced Systems*, Kuala Lumpur, Nov. 2007.  
doi:10.1109/ICIAS.2007.4658380

**C2.** R. Garhwal, **A. Halder**, and M. Sinha. An Adaptive Fuzzy State Noise Driven Extended Kalman filter for Real-time Orbit Determination. *58<sup>th</sup> International Astronautical Congress*, Hyderabad, Sep. 2007.

**C1.** S. Ghosh, **A. Halder**, and M. Sinha. Path Planning for A Fixed Wing Micro Air Vehicle in Fuzzy Quadtree Framework. *7<sup>th</sup> European Micro Air Vehicle Conference*, Toulouse, Sep. 2007.

PAPERS UNDER  
REVIEW/REVISION

**R.** G.A. Bondar, A. Eldesoukey, Y. Chen, and **A. Halder**. Nonlinear Non-Gaussian Density Steering with Input and Noise Channel Mismatch: Sinkhorn with Memory for Solving the Control-affine Schrödinger Bridge Problem.

**R.** T. Sial, and **A. Halder**. Symmetrizing Bregman Divergence on the Cone of Positive Definite Matrices: Which Mean to Use and Why.

**R.** A. Eldesoukey, Y. Chen, and **A. Halder**. A Generalized Sinkhorn Algorithm for Mean-Field Schrödinger Bridge.

**R.** H. Mahmood, **A. Halder**, and A. Akhtar. Schrödinger Bridge Over A Compact Connected Lie Group.

**R.** T. Sial, and **A. Halder**. Fixed Horizon Linear Quadratic Covariance Steering in Continuous Time with Hilbert-Schmidt Terminal Cost.

**R.** W. Wang, A.M.H. Teter, M. Arcak, and **A. Halder**. Set Invariance with Probability One for Controlled Diffusion: Score-based Approach.

**R.** A.M.H. Teter, W. Wang, S. Shivakumar, and **A. Halder**. Markov Kernels, Distances and Optimal Control: A Parable of Linear Quadratic Non-Gaussian Distribution Steering.

**R.** A.M.H. Teter, and **A. Halder**. On the Hopf-Cole Transform for Control-affine Schrödinger Bridge.

**R.** I. Nodozi, and **A. Halder**. Wasserstein Consensus ADMM.

**R.** W. Krichene, K.F. Caluya, and **A. Halder**. Global Convergence of Second-order Dynamics in Two-layer Neural Networks.

**R.** K.F. Caluya, and **A. Halder**. Finite Horizon Density Control for Static State Feedback Linearizable Systems.

POSTER  
PRESENTATIONS

**P4.** “Control of Large Scale Cyberphysical Systems”. *IEEE CDC*, Las Vegas, NV, Dec. 12, 2016.

**P3.** “Boolean Microgrid: A Theory of Operation for the Load Serving Entity”. *NSF CPS PI Meeting*, Arlington, VA, Oct. 31–Nov. 1, 2016.

**P2.** “Boolean Microgrid”. *NSF CPS PI Meeting*, Arlington, VA, Nov. 16–17, 2015.

**P1.** “A Control System Framework for Privacy Preserving Demand Response of Thermal Inertial Loads”. *Winedale Workshop*, Round Top, TX, Oct. 9, 2015.

## PROFESSIONAL ACTIVITIES

### Outreach

- *Creator and organizer* of “101 Topics in ML, Control and Optimization” with the help from Translational AI Center at Iowa State University. This is a research and educational outreach initiative. 101 = 1 topic, 0 slides, 1 whiteboard. Not a talk presenting one’s own/others’ latest research. Instead, high quality exposition of a topic/research area of contemporary interest in ML, Control and optimization.
- *Co-founder and co-instructor* for Cluster “Feedback Control with Applications to Robotics” in California State Summer School for Mathematics and Science (COSMOS), UC Santa Cruz, **Summer 2021, Summer 2022**. This is a 4-week summer program for high school scholars with demonstrated interest and achievement in mathematics and science.
- *Co-founder* of the NorCal Control Workshop, 2019 – present. This annual workshop brings together systems-control researchers from academia and industry in the Northern California region fostering collaboration and professional networking.

### Associate Editor

- *IEEE Transactions on Automatic Control*, Jan. 2026 – present.
- *ASME Journal of Dynamic Systems, Measurement, and Control*, Jan. 2025 – present.
- *Systems and Control Letters*, June 2022 – present.
- *IEEE Transactions on Aerospace and Electronic Systems*, Jan. 2019 – Dec. 2020.
- *IEEE Control Systems Society Conference Editorial Board*, June 2019 – July 2025.

### Reviewer for Papers

(The numbers below denote the number of *unique* papers reviewed, i.e., exclude the reviews performed for revised versions of the manuscripts in multiple rounds of reviews)

#### Journal (137)

- *IEEE Transactions on Automatic Control* (24)
- *IEEE Transactions on Information Theory* (2)
- *IEEE Transactions on Control Systems Technology* (1)
- *IEEE Transactions on Neural Networks and Learning Systems* (5)
- *IEEE Transactions on Artificial Intelligence* (1)
- *IEEE Transactions on Control of Network Systems* (1)
- *IEEE Control Systems Letters* (19)
- *IEEE Robotics and Automation Letters* (1)
- *IEEE Control Systems Magazine* (1)
- *IEEE Transactions on Power Systems* (3)
- *IEEE Transactions on Smart Grid* (8)
- *IEEE Internet of Things Journal* (4)
- *IEEE Transactions on Computational Imaging* (1)
- *SIAM Journal on Control and Optimization* (5)
- *SIAM Journal on Mathematics of Data Science* (1)
- *SIAM Review* (1)
- *Automatica* (14)
- *International Journal of Robust and Nonlinear Control* (2)
- *International Journal of Control* (1)
- *IMA Journal of Mathematical Control and Information* (1)
- *Journal of Optimization Theory and Applications* (2)
- *Optimal Control Applications and Methods* (2)

- *IET Control Theory & Applications* (1)
- *AIAA Journal of Guidance, Control, and Dynamics* (3)
- *ASME Journal on Dynamic Systems, Measurement and Control* (24)
- *Proceedings of the Royal Society A* (1)
- *The European Physical Journal Plus* (1)
- *Advances in Space Research* (1)
- *Electric Power Systems Research* (3)
- *Energy Science & Engineering* (1)
- *Energies* (2)

Conference (90)

- *IEEE Conference on Decision and Control* 2025 (3)
- *IEEE Conference on Decision and Control* 2024 (2)
- *American Control Conference* 2024 (2)
- *IEEE Conference on Decision and Control* 2023 (3)
- *International Conference on Machine Learning* 2023 (1)
- *European Control Conference* 2023 (1)
- *26th International Conference on Artificial Intelligence and Statistics* 2023 (3)
- *American Control Conference* 2023 (2)
- *Conference on Neural Information Processing Systems* 2022 (5)
- *4th IFAC Workshop on Cyber-Physical & Human-Systems* 2022 (2)
- *IEEE Conference on Decision and Control* 2022 (2)
- *International Conference on Machine Learning* 2022 (2)
- *Mathematical Theory of Networks and Systems* 2022 (1)
- *American Control Conference* 2022 (3)
- *IEEE Conference on Decision and Control* 2021 (2)
- *American Control Conference* 2021 (3)
- *IEEE Conference on Decision and Control* 2020 (3)
- *Mathematical Theory of Networks and Systems* 2020 (1)
- *American Control Conference* 2020 (2)
- *IEEE Conference on Decision and Control* 2019 (3)
- *American Control Conference* 2019 (2)
- *Indian Control Conference* 2019 (2)
- *IEEE Conference on Decision and Control* 2018 (3)
- *Mathematical Theory of Networks and Systems* 2018 (1)
- *American Control Conference* 2018 (3)
- *IEEE Conference on Decision and Control* 2017 (3)
- *IEEE Power & Energy Society General Meeting* 2017 (1)
- *American Control Conference* 2017 (1)
- *IEEE Conference on Decision and Control* 2016 (3)
- *American Control Conference* 2016 (3)
- *American Control Conference* 2015 (4)
- *IEEE Multi-conference on Systems and Control* 2014 (1)
- *IEEE Conference on Decision and Control* 2014 (1)
- *American Control Conference* 2014 (3)
- *IEEE Conference on Decision and Control* 2013 (1)
- *American Control Conference* 2013 (4)
- *American Control Conference* 2012 (2)
- *ASME Dynamic Systems and Control Conference* 2012 (1)
- *IEEE Conference on Decision and Control* 2011 (1)
- *IEEE Conference on Robotics and Automation* 2010 (2)
- *American Control Conference* 2009 (1)
- *IEEE International Conference on Intelligent and Advanced Systems* 2007 (1)

### Invited Reviewer for Books

- Springer Mathematics Series
- IEEE Control Systems Magazine

### Invited Reviewer for Grant Proposals/Panels

- NSF
- University Grants Academy, San José State University

### Conference/Workshop Organization

- *Member* of the IEEE CDC 2025 Best Student Paper Award (BSPA) Committee.
- *Member* of the IEEE CDC 2024 Best Student Paper Award (BSPA) Committee.
- *Co-organizer (with M. Arcak, H. Litz, L. Pham, R. Sanfelice, and M. Zamani)*, 4th Workshop on “Computation-Aware Algorithmic Design for Cyber-Physical Systems” at 2024 CPS-IoT week, Hong Kong, People’s Republic of China, May 13, 2024.
- *Chair* for Session: “Uncertain Systems I”, co-Chair for Session: “Stochastic Systems and Control I”, American Control Conference 2024.
- *Member* of the IEEE CDC 2023 Best Student Paper Award (BSPA) Committee.
- *Chair* for Sessions: “Learning”, “Learning and Stochastic Optimal Control”, American Control Conference 2023.
- *Co-organizer (with A. Mesbah)*, Invited Session on “Learning and Stochastic Optimal Control”, American Control Conference 2023.
- *Co-organizer (with M. Arcak, H. Litz, L. Pham, R. Sanfelice, and M. Zamani)*, 3rd Workshop on “Computation-Aware Algorithmic Design for Cyber-Physical Systems” at 2023 CPS-IoT week, San Antonio, Texas, May 9, 2023.
- *International Program Committee Member* of the 4th IFAC Workshop on Cyber-Physical & Human-Systems (CPHS), December 1-2, 2022.
- *Co-organizer (with M. Arcak, H. Litz, L. Pham, R. Sanfelice, and M. Zamani)*, 2nd Workshop on “Computation-Aware Algorithmic Design for Cyber-Physical Systems” at 2022 CPS-IoT week, May 3, 2022.
- *Co-organizer (with M. Chertkov, and M. Korkali)*, Full day workshop: “Uncertainty Management in Power System Dynamics” at IEEE Conference on Decision and Control 2021.
- *Chair* for Session: “Stochastic Systems”, American Control Conference 2021.
- *Co-organizer (with E. Bakolas, Y. Chen, and P. Tsiotras)*, Full day workshop: “Control of Distributions: Theory and Applications” at 2021 American Control Conference, May 24, 2021.
- *Co-organizer (with M. Arcak, H. Litz, L. Pham, R. Sanfelice, and M. Zamani)*, 1st Workshop on “Computation-Aware Algorithmic Design for Cyber-Physical Systems” at 2021 CPS-IoT week, May 18, 2021.
- *Chair* for Session: “Uncertain Systems II”; *co-Chair* for Session: “Stochastic Systems”, American Control Conference 2020.
- *Co-organizer (with K. Sreenath, M. Arcak, and R. Sanfelice)*, CITRIS/CPAR Control Theory and Automation Symposium and 2<sup>nd</sup> Norcal Control Workshop, University of California Berkeley, April 26, 2019.
- *Chair* for Session: “Optimization Algorithms III”; *co-Chair* for Session: “Markov Processes II”, American Control Conference 2019.
- *Co-Chair*, Session: “Computational Methods II”, IEEE Conference on Decision and Control 2018.
- *Chair*, Session: “Filtering”, American Control Conference 2018.
- *Organizer (with R. Sanfelice, K. Goldberg, and R. Berenstein)*, CITRIS/CPAR Control Theory and Automation Symposium and 1<sup>st</sup> Norcal Control Workshop, University of California Santa Cruz, April 27, 2018.

- *Member* of the IEEE Control Systems Society (CSS) Technical Committee on “Systems with Uncertainty” (Jan 2018 – current).
- *Organizer (with P.R. Kumar and L. Xie)*, Invited Session: “Recent Advances in Control of Thermal Inertial Loads and DC Microgrid Stability”, American Control Conference 2017.
- *Co-Chair*, Session: “Modeling”, IEEE Conference on Decision and Control 2014.

#### **Selected University Service**

- *ABET Program Review Committee*, Department of Aerospace Engineering, Iowa State University, 2023–current.
- *Best Dissertation Award Committee*, Cyber-Physical Systems Research Center, UC Santa Cruz, 2022–23.
- *Faculty Hiring Committee on Scientific Machine Learning*, Department of Applied Mathematics, UC Santa Cruz, 2022–23.
- *Senate Committee Member for Committee on Library and Scholarly Communication*, UC Santa Cruz, 2022–23.
- *Senate Committee Member for Committee of Research*, UC Santa Cruz, 2019–20.
- *Member of Website Committee*, Baskin School of Engineering, UC Santa Cruz, 2019–20.
- *Member of Graduate Committee*, Department of Applied Mathematics, UC Santa Cruz, 2018–23.

#### **AWARDS**

##### **Research Awards**

- O. Hugo Schuck Best Application Paper Award  
For paper: *A Physics-informed Deep Learning Approach for Minimum Effort Stochastic Control of Colloidal Self-Assembly*, American Automatic Control Council, 2024.
- Outstanding Doctoral Student Award  
*Department of Aerospace Engineering*, Texas A&M University, 2014.
- Best Presentation in Session Award  
*Session: ‘Filtering’*, *American Control Conference*, Washington, D.C., 2013.
- Best Thesis Award (Dual Degree)  
*Development of An Autonomous Reconfigurable UAV*  
Department of Aerospace Engineering, IIT Kharagpur, INDIA 2008.

##### **Mentee Awards**

- Alexis M.H. Teter  
*Springer Outstanding Ph.D. Thesis Award (2026)*  
*Achievement Rewards for College Scientists (ARCS) Foundation Fellowship (2024-25)*
- Iman Nodozi  
*O. Hugo Schuck Best Application Paper Award (2024)*, *American Automatic Control Council*  
*Baskin School of Engineering Dissertation Year Fellowship (2023-24)*  
*UC Santa Cruz Regent’s Fellowship (2019-20)*
- Georgiy A. Bondar  
*Dean’s Fellowship (2022-23)*



- Shadi Haddad  
*Applied Mathematics Research Award (2021-22)*  
*UC Santa Cruz Chancellor's Fellowship (2019-20)*

#### Travel Awards

- Institute of Mathematics and its Applications (IMA) Travel Support Award  
*Workshop on Control at Large Scales: Energy Markets and Responsive Grids*,  
IMA Thematic Year on Control Theory and its Applications, Minneapolis, 2016.
- IEEE Control Systems Society Student Travel Award  
*American Control Conference*, Portland, 2014.
- IEEE Control Systems Society Student Travel Award  
*American Control Conference*, Washington, D.C., 2013.
- IEEE Control Systems Society Student Travel Award  
*51<sup>st</sup> IEEE Conference on Decision and Control*, Maui, 2012.

#### INVITED TALKS (EXCLUDING CONFERENCE PAPER TALKS)

- T61.** Invited Speaker at the ControlX Seminar, University of Washington, Seattle, WA, April 06, 2026.
- T60.** Invited Speaker at the Mechanical and Aerospace Engineering Seminar Series, University of California, Irvine, CA, March 06, 2026.
- T59.** Invited Speaker at the Computational and Applied Mathematics Seminar, University of Arizona, Tuscon, AZ, October 02, 2025.
- T58.** Invited Speaker at the Decision Superiority Seminar, Lawrence Livermore National Laboratory, Livermore, CA, March 03, 2025.
- T57.** Invited Speaker at the Decision Superiority Seminar, Lawrence Livermore National Laboratory, Livermore, CA, February 24, 2025.
- T56.** Invited Speaker at the Mechanical Colloquium, New Jersey Institute of Technology, NJ, February 12, 2025.
- T55.** Invited Speaker at the Computational and Applied Math Seminar, Department of Mathematics, Iowa State University, IA, February 10, 2025.
- T54.** Invited Speaker at the Decision Superiority Seminar, Lawrence Livermore National Laboratory, Livermore, CA, December 05, 2024.
- T53.** Invited Speaker at the 5th Workshop on Physics Informed Machine Learning, Center for Nonlinear Studies at Los Alamos National Laboratory, Los Alamos, NM, October 17, 2024.
- T52.** Invited Speaker at the 60th Annual Allerton Conference Session: Duality in Control, Inference, and Learning, National Center for Supercomputing Applications, Urbana, IL, September 25, 2024.
- T51.** Invited Speaker at the Department of Mathematics, University of Iowa, Iowa City, IA, September 03, 2024.
- T50.** Invited Speaker at the Learning, Computation and Control Seminar, Imperial College, London, June 06, 2024.
- T49.** Invited Speaker at the Level Set Seminar, Department of Mathematics, University of California Los Angeles, May 13, 2024.

- T48.** Invited Speaker at the Guidance, Navigation and Control Meeting, Department of Aerospace Engineering, Iowa State University, Ames, April 02, 2024.
- T47.** Invited Speaker at the Analysis and Probability Seminar, Iowa State University, Ames, March 06, 2024.
- T46.** Invited Speaker at the CPS Frontier Workshop, University of California Berkeley, CA, November 03, 2023.
- T45.** Invited Speaker at the Decision and Control Seminar, University of Illinois at Urbana-Champaign, IL, November 01, 2023.
- T44.** Invited Speaker at the Mathematics and Deep Learning Collective, Iowa State University, Ames, September 27, 2023.
- T43.** Invited Speaker at the Translational AI Center Seminar, Iowa State University, Ames, September 22, 2023.
- T42.** Invited Guest Lecturer in Math 577 course on “Stochastic Control and Learning” by M. Chertkov, Special Applied Mathematics Seminar, University of Arizona, Tuscon, September 18, 2023.
- T41.** Invited Speaker at the Mini-symposium on Mean Field Games and Optimal Transport with Applications in Data Science and Biology, 10th International Congress on Industrial and Applied Mathematics (ICIAM), Tokyo, JAPAN, August 21, 2023.
- T40.** Invited Speaker at the SIAM Mini-symposium on Optimal Transport: Theory and Applications in Systems and Control, SIAM Conference on Control and Its Applications, Philadelphia, PA, July 25, 2023.
- T39.** Invited Speaker at the IEEE Control Systems Society Chapter Invited Talk, Indian Institute of Technology Kharagpur, INDIA, January 16, 2023.
- T38.** Invited Speaker at the Dept. of Aerospace Engineering, Iowa State University, Ames, November 17, 2022.
- T37.** Invited Speaker at the Dept. of Applied Mathematics, University of California, Santa Cruz, November 07, 2022.
- T36.** Invited Speaker at the Dept. of Mechanical Engineering, University of Alabama, Tuscaloosa, November 04, 2022.
- T35.** Invited Speaker at the Palo Alto Research Center, June 14, 2022.
- T34.** Invited Speaker at the 2022 NSF AMPS PIs Workshop, George Mason University, May 26, 2022.
- T33.** Invited Speaker at Yahoo! Research, March 30, 2022.
- T32.** Invited Speaker at the Optimal Transport and Mean Field Games Seminar, University of South Carolina, January 26, 2022.
- T31.** Invited Speaker at the Technical Design Review, Ford Research and Advanced Engineering, January 20, 2022.
- T30.** Invited Speaker at the Two Day Workshop on Uncertainty Management in Power System Dynamics, 60th IEEE conference on Decision and Control, December 12, 2021.
- T29.** Discovery Lecture at the 2021 California State Summer School for Mathematics and Science (COSMOS), University of California Santa Cruz, CA, July 19, 2021.

- T28.** Invited Speaker at the Full Day Workshop on Control of Distributions: Theory and Applications, 2021 American Control Conference, May 24, 2021.
- T27.** Invited Speaker at the 1st Workshop on Computation-Aware Algorithmic Design for Cyber-Physical systems, 2021 CPS-IoT week, May 18, 2021.
- T26.** Invited Speaker at the 2020 NSF AMPS PIs Workshop, Virtual event, November 19, 2020.
- T25.** Invited Speaker at the Controls, Autonomy and Robotics Seminar, University of Texas, Austin, November 18, 2020.
- T24.** Invited Speaker at the Probabilistics Seminar Series, GE Research, May 27, 2020.
- T23.** Invited Speaker at the SIAM mini-symposium on “Optimal Control Methods for Nonlinear Filtering and Data Assimilation”, SIAM Conference on Uncertainty Quantification (UQ), Munich, Germany, March 26, 2020.  
[Canceled due to COVID-19]
- T22.** Applied Mathematics Seminar, Naval Postgraduate School, Monterey, CA, February 25, 2020.
- T21.** Invited Speaker at the “Uncertainty Synthesis” Workshop (half-day), 2019 CDC, Nice, France, December 10, 2019.
- T20.** Invited Speaker at the 2019 Bay Area Robotics Symposium, University of California, Berkeley, CA, November 15, 2019.
- T19.** Google Research Invited Seminar, Mountain View, CA, October 31, 2019.
- T18.** 2019 NSF AMPS PIs Workshop, George Washington University, Washington DC, October 23, 2019.
- T17.** Applied Mathematics Seminar, University of California Santa Cruz, CA, October 07, 2019.
- T16.** Center for Control, Dynamical Systems, and Computation Seminar, University of California Santa Barbara, CA, October 04, 2019.
- T15.** Electrical and Computer Engineering Seminar, University of California Santa Cruz, CA, May 20, 2019.
- T14.** Mathematics/Statistics colloquium, San Jose State University, San Jose, CA, October 10, 2018.
- T13.** Center for Information Technology in the Interest of Society (CITRIS) “People and Robots” and “Design of Robotics and Embedded systems, Analysis, and Modeling” (DREAMS) Seminar, University of California, Berkeley, CA, February 12, 2018.
- T12.** Special Seminar at Jack Baskin School of Engineering, University of California Santa Cruz, CA, December 4, 2017.
- T11.** Second Annual Center for Research in Open Source Software (CROSS) Research Symposium, University of California Santa Cruz, CA, October 4, 2017.
- T10.** 32<sup>nd</sup> Southern California Control Workshop, Caltech, CA, April 21, 2017.
- T9.** Department of Aerospace Engineering, Mississippi State University, MS, April 13, 2017.
- T8.** Department of Mechanical and Aerospace Engineering, Syracuse University, NY, March 31, 2017.

- T7.** Department of Mechanical Engineering, University of Texas at Dallas, TX, March 20, 2017.
- T6.** Department of Applied Mathematics and Statistics, University of California Santa Cruz, CA, January 27, 2017.
- T5.** Comverge Inc., Denver, CO, December 5, 2016.
- T4.** Workshop on Architecture and Economics of the Future Grid, Texas A&M University, College Station, TX, November 3, 2016.
- T3.** Electric Power and Power Electronics Institute Seminar, Department of Electrical and Computer Engineering, Texas A&M University, College Station, TX, October 26, 2015.
- T2.** Schlumberger-Doll Research Center, Cambridge, MA, July 8, 2014.
- T1.** Department of Mechanical and Aerospace Engineering, University of Florida, Gainesville, FL, May 15, 2012.

TEACHING  
EXPERIENCE

**Iowa State University**, Ames, Iowa USA

*Instructor for AERE 5740: Optimal Control*  
**Spring 2024, Spring 2026**

- Graduate level, class size: 16 (S24), 11 (S26).
- Calculus of variations and optimal control.

*Instructor for AERE 331: Flight Control Systems I*  
**Fall 2023, Fall 2024**

- Undergraduate level, class size: 67 (F23), 71 (F24).
- Introduction to control systems with focus on flight control.

**University of California, Santa Cruz**, Santa Cruz, California USA

*Instructor for AM/AMS 20: Mathematical Methods for Engineers II*  
**Spring 2018, Spring 2019, Spring 2020**

- Undergraduate level, class size: 131 (S18), 167 (S19), 161 (S20).
- Ordinary differential equations.
- Course website: [AMS20-S18](#), [AMS20-S19](#), [AM20-S20](#)

*Instructor for AM 147: Computational Methods and Applications*  
**Winter 2020, Winter 2021, Winter 2022, Winter 2023**

- Undergraduate level, class size: 143 (W20), 124 (W21), 131 (W22), 138 (W23).
- Numerical methods for solving scientific and engineering problems.
- Course website: [AM147-W20](#), [AM147-W21](#), [AM147-W22](#), [AM147-W23](#)

*Instructor for AM/AMS 229: Convex Optimization*  
**Fall 2018, Fall 2020, Fall 2022**

- Graduate level, class size: 12 (F18), 33 (F20), 27 (F22).
- Recognizing, analyzing and transforming convex optimization problems with focus on solving the same using `cvx` in MATLAB, Python or Julia.
- Course website: [AMS229-F18](#), [AM229-F20](#), [AM229-F22](#)

*Instructor for AM/AMS 232: Applied Optimal Control*  
**Spring 2019, Spring 2021**

- Graduate level, class size: 15 (S19), 9 (S21).
- Deterministic and stochastic optimal control.
- Course website: [AMS232-S19](#), [AM232-S21](#)

*Instructor for AM/AMS 231: Nonlinear Control Theory*

**Winter 2018, Spring 2020, Spring 2022**

- Graduate level, class size: 16 (W18), 11 (S20), 6 (S22).
- Control of finite dimensional nonlinear systems.
- Course website: [AMS231-W18](#), [AM231-S20](#), [AM231-S22](#)

*Instructor for AM/AMS 280B: Seminar in Applied Mathematical Modeling*

**Winter 2019, Fall 2019, Winter 2020, Winter 2021, Fall 2021, Winter 2022, Fall 2022**

- Graduate level departmental seminar during the academic quarters: Fall, Winter and Spring.
- Invitation and hosting of external seminar speakers in research areas of interest in applied mathematics.

**University of California, Irvine**, Irvine, California USA

*Co-lecturer for MAE 295: Networks and Control*

**Spring 2017**

- Graduate level course.
- Delivered in-class lectures on dynamics and control of multi-agent systems over networks.

**Texas A&M University**, College Station, Texas USA

*Teaching Assistant for AERO 320: Numerical Methods*

**Fall 2013**

- Junior level undergraduate course.
- Designed and graded homeworks and tests.
- Designed lab assignments and conducted lab sessions for implementing the numerical methods in C++.
- Held help sessions.
- Course material: [abhishekhalder.org/Aero320Fall2013](http://abhishekhalder.org/Aero320Fall2013)

*Grader for ENGR 111: Foundations of Engineering*

**Fall 2009**

- Freshman level undergraduate course.
- Graded weekly assignments on engineering mechanics and statistics.
- Held weekly help sessions.

**Indian Institute of Technology Kharagpur**, West Bengal INDIA

*Instructor for AE21008: Introduction to Flight Vehicle Controls*

**Spring 2008**

- Sophomore level undergraduate course.
- Delivered in-class lectures on the basics of feedback control systems, block diagrams, dynamic system modeling and response, designing PID controller, root-locus design, frequency response design, state space design.

MENTORING AND  
ADVISING

*Postdoctoral Scholars*

- Asmaa Eldesoukey (Aerospace Engineering, Iowa State University). March 2026 – present.
- Wenqing Wang (Aerospace Engineering, Iowa State University). Feb. 2024 – Feb. 2025. Next position: Research Engineer, Shenzhen Xinkailai Technology Co., Ltd., China.
- Sachin Shivakumar (Aerospace Engineering, Iowa State University). May 2024 – Jan. 2025. Next position: Postdoctoral Scholar at Theoretical Division, Los Alamos National Lab, NM.

#### *Doctoral Students*

- Primary supervisor: Chiran Cherian (Aerospace Engineering, Iowa State University). Summer 2026 – present.
- Primary supervisor: Md Zulfiqur Haider (Co-major in Mathematics and Aerospace Engineering, Iowa State University). Summer 2026 – present.
- Primary supervisor: Ripon Sarker (Aerospace Engineering, Iowa State University). Spring 2026 – present.
- Primary supervisor: Georgiy Bondar (Applied Mathematics, UC Santa Cruz). Fall 2022 – present. [**Winner of 2022 Dean’s Fellowship**]
- Primary supervisor: Alexis Teter (Applied Mathematics, UC Santa Cruz). Fall 2021 – Fall 2025. [**Winner of 2026 Springer Outstanding Ph.D. Thesis Award, 2024-25 Achievement Rewards for College Scientists (ARCS) Foundation Fellowship**], Next position: Upper School Teacher in Mathematics, Computer and Data Science, Crystal Springs Uplands School, Hillsborough, CA.  
**Ph.D. Dissertation:** *Contractions and Reactions in Schrödinger Bridges.*
- Primary supervisor: Iman Nodozi (Electrical and Computer Engineering, UC Santa Cruz). Summer 2021 – Summer 2024. [**Winner of 2024 O. Hugo Schuck Best Application Paper Award, 2023-24 Baskin School of Engineering Dissertation Year Fellowship, 2018-19 Regent’s Fellowship**], Next position: Senior Application Engineer, Onsemi, CA.  
**Ph.D. Dissertation:** *Measure-valued Proximal Recursions for Learning and Control.*
- Primary supervisor: Shadi Haddad (Applied Mathematics, UC Santa Cruz). Fall 2019 – Spring 2024. [**Winner of 2022 Applied Mathematics Research Award, 2018-19 Chancellor’s Fellowship**]  
**Ph.D. Dissertation:** *A Geometric Approach for Learning Reach Sets.*
- Primary supervisor: Kenneth Caluya (Applied Mathematics, UC Santa Cruz). Fall 2017 – Fall 2022.  
**Ph.D. Dissertation:** *Generalized Gradient Flows for Density Prediction, Control and Learning.*
- Ph.D. Committee Member: Ashley Behrendt (Aerospace Engineering, Iowa State University). Spring 2026 – present.
- Ph.D. Committee Member: Odai Bani Hani (Aerospace Engineering, Iowa State University). Spring 2026 – present.
- Ph.D. Committee Member: Alec Rosentrater (Aerospace Engineering, Iowa State University). Summer 2025 – present.
- External Ph.D. Committee Member: Hamidreza Behjoo (Department of Mathematics, University of Arizona, Tucson). Fall 2024 – present.
- Ph.D. Committee Member: Negrete Fernandez De Aimar (Aerospace Engineering, Iowa State University). Spring 2025 – present.
- Ph.D. Committee Member: Aadit Farhan Khan (Mechanical Engineering, Iowa State University). Fall 2024 – present.
- Ph.D. Committee Member: Alex Perrucci (Aerospace Engineering, Iowa State University). Fall 2024 – Spring 2025.

- Ph.D. Committee Member: Muhammad Sarwar (Electrical Engineering, Iowa State University). Fall 2024 – Summer 2025.
- Ph.D. Committee Member: Soumya Roy (Electrical Engineering, Iowa State University). Spring 2024 – Spring 2025.
- External Ph.D. Committee Member: Isin Mert Balci (Aerospace Engineering and Engineering Mechanics, UT Austin). Fall 2023 – Fall 2024.
- Ph.D. Committee Member: Santiago Jimenez Leudo (Electrical and Computer Engineering, UC Santa Cruz). Fall 2021 – Fall 2024.
- Ph.D. Committee Member: Abram Rodgers (Applied Mathematics, UC Santa Cruz). Spring 2021 – Spring 2023.
- Ph.D. Committee Member: Tenavi Nakamura-Zimmerer (Applied Mathematics, UC Santa Cruz). Fall 2019 – Spring 2022.
- Ph.D. Committee Member: Marcello Guarro (Electrical and Computer Engineering, UC Santa Cruz). Fall 2019 – Summer 2021.
- Ph.D. Committee Member: Dawn Hustig-Schultz (Electrical and Computer Engineering, UC Santa Cruz). Fall 2018 – Spring 2023.
- Ph.D. Committee Member: Richard Shaffer (Applied Mathematics, UC Santa Cruz). Fall 2017 – Winter 2018.

#### *Masters Students*

- Primary supervisor: Tushar Sial (Aerospace Engineering, Iowa State University). Fall 2024 – present.
- Primary supervisor: Charlie Yan (Electrical and Computer Engineering, UC Santa Cruz). Summer 2022 – Spring 2023. Next position: Robotics Engineer, Meta, CA.  
**M.S. Thesis:** *Neural Schrödinger Bridge with Sinkhorn Losses.*
- Primary supervisor: Qingyuan Cui (Applied Mathematics, UC Santa Cruz). Summer 2020 – Summer 2021. Next position: Graduate student in Data Science at John Hopkins University, MD.  
**M.S. Thesis:** *Graph Curvature for COVID-19 Network Risk Analytics.*  
Primary supervisor: Lia Gianfortone (Applied Mathematics, UC Santa Cruz). Fall 2017 – Summer 2018.  
**M.S. Thesis:** *Ellipsoidal Algorithm for Fast Computation of Reachable Tubes.*
- M.S. Thesis Committee Member: Andrea Bonacci (Aerospace Engineering, Iowa State University). Spring 2021.
- M.S. Thesis Committee Member: Harsh Bhakta (Computer Science and Engineering, UC Santa Cruz). Spring 2021.
- M.S. Thesis Committee Member: David Kooi (Electrical and Computer Engineering, UC Santa Cruz). Spring 2020 – Winter 2021.
- M.S. Thesis Committee Member: Adam Ames (Electrical and Computer Engineering, UC Santa Cruz). Spring 2020.
- M.S. Thesis Committee Member: Wuyuan Chen (Electrical and Computer Engineering, UC Santa Cruz). Fall 2018.
- M.S. Thesis Committee Member: Yegeta Zeleke (Electrical and Computer Engineering, UC Santa Cruz). Fall 2018.
- M.S. Thesis Committee Member: Marcello Guarro (Electrical and Computer Engineering, UC Santa Cruz). Spring 2018 – Summer 2019.

#### *Undergraduate Students*

- Primary supervisor: Karthik Sivaramakrishnan (Mathematics, UC Santa Cruz). Summer 2020 – Fall 2020.  
**Senior Thesis:** *Ollivier-Ricci Curvature for Directed Weighted Graphs.*

*High school Students*

- Pranav Eranki (Cupertino High School). Winter 2020 – Fall 2020.

PROFESSIONAL  
MEMBERSHIP

Senior member, IEEE  
IEEE Control Systems Society (CSS)  
Member, International Federation of Automatic Control (IFAC)  
Member, Society of Industrial and Applied Mathematics (SIAM)  
Member, American Society of Mechanical Engineers (ASME)